

Assignment #4

Reliable and Trustworthy Artificial Intelligence

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1 Exploration of the α, β -CROWN Models Directory

Models. The `complete_verifier/models/` directory contains pretrained networks organized by benchmark category. Unlike Marabou, which stores all models in a single `nnet/` folder in the `.nnet` format, α, β -CROWN provides models in ONNX and PyTorch formats across a wider range of architectures: (1) **MNIST** — small MLP and CNN models for initial experiments; (2) **CIFAR-10/100 and TinyImageNet** — deeper CNN and ResNet models (e.g., `cifar_resnet_2b`) for high-dimensional inputs; (3) **ACASXu** — the same low-dimensional safety-critical networks also found in Marabou’s resources, enabling direct comparison between verifiers; (4) **NN4sys, ML4ACOPF** — domain-specific models for systems and power grid applications.

YAML Configuration Files.

All verification parameters are defined in YAML files located in `complete_verifier/exp_configs/`. A single YAML file covers model path, dataset, verification property, and solver hyperparameters (e.g., batch size, PGD attack steps, branch-and-bound timeout). This is the key organizational difference from Marabou: whereas Marabou requires manual Python API calls to encode each property, α, β -CROWN centralizes all configuration in a single human-readable file, making experiments easier to reproduce and share.

Comparison with Marabou.

Marabou’s `resources/` directory focuses on a small set of safety-critical benchmarks (`acasxu`, `coav`, `twin`) in the legacy `.nnet` format, with property files specified separately as plain text. α, β -CROWN covers a broader range of benchmarks, supports modern ONNX/PyTorch formats, and integrates GPU-accelerated verification natively.

2 Model, Dataset, and Verification Property

Model. We reused the same small fully-connected network from Assignment #3:

$$\text{Input}(784) \rightarrow \text{Linear}(32) \rightarrow \text{ReLU} \rightarrow \text{Linear}(16) \rightarrow \text{ReLU} \rightarrow \text{Linear}(10)$$

exported as `mnist_fc.onnx` (ONNX opset 11). This model is not included in α, β -CROWN’s model directory. Reusing the same model as Assignment #3 enables a direct performance comparison between the two verifiers. The model achieves 95.93% test accuracy on MNIST.

Dataset. MNIST handwritten digit dataset (10,000 test images, 28×28 grayscale, pixel values in $[0, 1]$). We verified the first 10 test samples.

Verification property. Local ℓ_∞ robustness with $\varepsilon = 0.01$: for each test image $\mathbf{x}$, verify that all $\mathbf{x}'$ satisfying $\|\mathbf{x}' - \mathbf{x}\|_\infty \leq 0.01$ are classified as the same digit.

Configuration. The YAML configuration specified `bound_prop_method: alpha-crown`, `batch_size: 256`, `timeout: 60s`, and `device: cpu`. No modifications to the ONNX model were required; α, β -CROWN loaded it directly via `onnx2pytorch`.

Installation issues encountered. Two issues arose during setup. First, a version conflict between the conda-installed `graphviz` (0.21) and a previously pip-installed version caused an `AttributeError` in `graphviz.backend`; resolved by force-reinstalling `graphviz==0.21` via pip after removing the conda version. Second, the built-in MNIST data loader searched for data in `complete_verifier/datasets/MNIST/raw/` rather than the default torchvision path; resolved by creating a symbolic link to the pre-downloaded MNIST files from Assignment #3.

3 Results and Interpretation

Table 1 summarizes the verification results for 10 test samples.

Outcome	Count	Indices
safe-incomplete (verified)	9	0, 1, 2, 3, 4, 5, 6, 7, 9
unsafe-pgd (falsified)	1	8
timeout	0	—
Verified accuracy	90%	

Table 1: Verification results: MNIST FC model, $\varepsilon = 0.01$, 10 samples.

Mean verification time per sample: **0.499 s** (max: 1.537 s).

Interpretation. Nine samples were labeled *safe-incomplete*: α -CROWN’s linear relaxation was sufficient to certify robustness without invoking branch-and-bound, indicating tight output bounds. Sample index 8 was labeled *unsafe-pgd*: a PGD adversarial attack found a counterexample within the ε -ball, meaning the model fails the robustness property at that point. This is consistent with the model’s imperfect test accuracy — some samples lie near the decision boundary and are susceptible to small perturbations.

4 Comparison: α, β -CROWN vs. Marabou

Both tools were applied to the same model (`mnist_fc.onnx`) with $\varepsilon = 0.01$. Table 2 summarizes the key differences.

	Marabou	α, β -CROWN
Verification method	SMT + Simplex	Bound Propagation + BnB
GPU support	No	Yes (CUDA)
Model format	.nnet, .onnx, .pb	ONNX, PyTorch
Property spec.	Python API	YAML config file
Samples verified	1 image (digit 7)	10 images
Result	UNSAT (all 9 pairs)	9 safe, 1 falsified
Total time	~ 0.3 s	~ 5.0 s (10 samples)
Time per sample	~ 0.03 s/pair	~ 0.5 s/sample

Table 2: Comparison of Marabou and α, β -CROWN on the same MNIST FC model.

Marabou’s SMT-based approach is faster per query on this small model, but α, β -CROWN provides richer results by simultaneously running PGD attacks to detect non-robust samples. Marabou returned formal UNSAT guarantees (complete verification), while α, β -CROWN returned *safe-incomplete* results — sound but not complete — for the 9 safe samples, meaning the bounds were sufficient to certify safety without full BnB.

5 Discussion: Strengths and Limitations of α, β -CROWN

Strengths. α, β -CROWN scales to significantly larger networks than Marabou, supporting CNN and ResNet architectures that would be intractable for SMT-based methods. GPU acceleration via CUDA enables massive parallelism in branch-and-bound (batch processing of subdomains), making it the state-of-the-art verifier across five consecutive VNN-COMP competitions (2021–2025). The YAML-based configuration system makes experiments easy to reproduce, and the built-in PGD attacker provides counterexamples when robustness fails without additional tooling.

Limitations. Installation proved non-trivial: version conflicts between conda and pip packages (`graphviz`) and undocumented data path conventions (`datasets/` vs. `data/`) required manual debugging. The *safe-incomplete* label, while sound, does not provide a formal UNSAT proof equivalent to Marabou’s complete verification. Furthermore, α, β -CROWN requires CUDA for optimal performance; running on CPU significantly increases verification time.